

Q1.

In this question use $g = 9.8 \text{ m s}^{-2}$.

A composite body consists of a uniform rod, AB , and a particle.

The rod has length 4 metres and mass 22.5 kilograms.

The particle, P , has mass 20 kilograms and is placed on the rod at a distance of 0.3 metres from B , as shown in the diagram.



- (a) Find the distance of the centre of mass of the body from A .

(2)

- (b) The body rests in equilibrium in a horizontal position on two supports, C and D . The support C is 0.5 metres from A and the support D is 1 metre from B .

Find the magnitude of the forces exerted on the body by the supports.

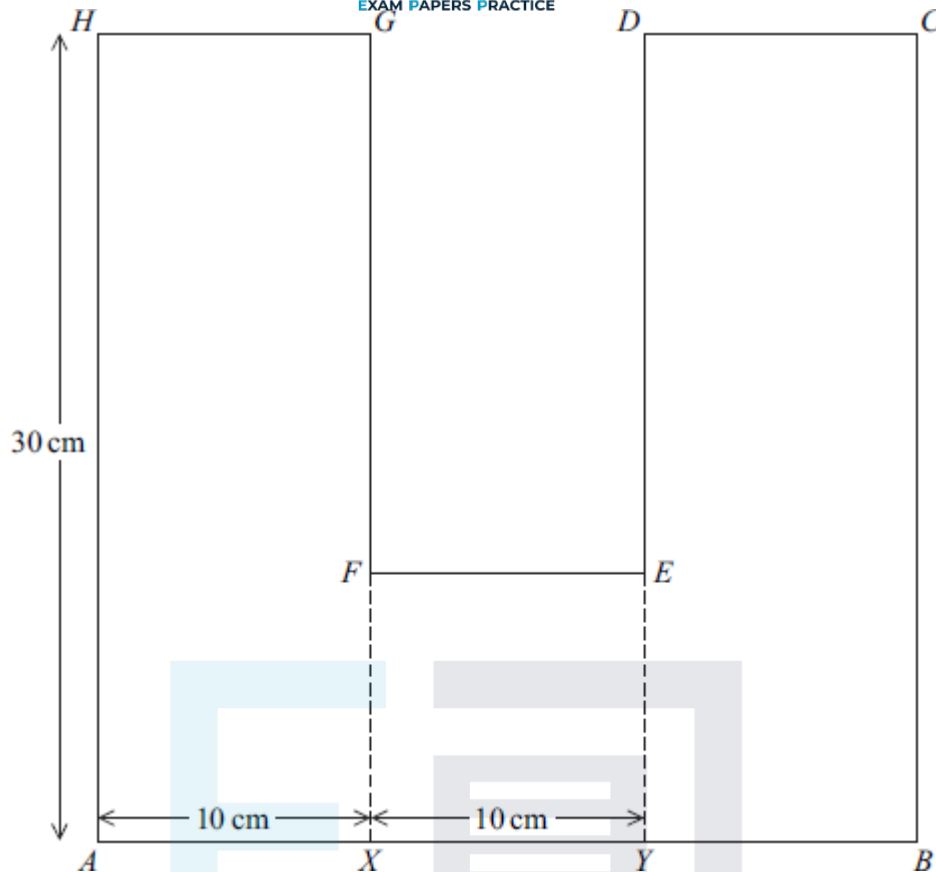
(4)

(Total 6 marks)

Q2.

The diagram shows a uniform lamina which is in the shape of two identical rectangles $AXGH$ and $YBCD$ and a square $XYEF$, arranged as shown.

The length of AX is 10 cm, the length of XY is 10 cm and the length of AH is 30 cm.



- (a) Explain why the centre of mass of the lamina is 15 cm from AH .

(1)

- (b) Find the distance of the centre of mass of the lamina from AB .

(3)

- (c) The lamina is freely suspended from the point H .

Find, to the nearest degree, the angle between HG and the horizontal when the lamina is in equilibrium.

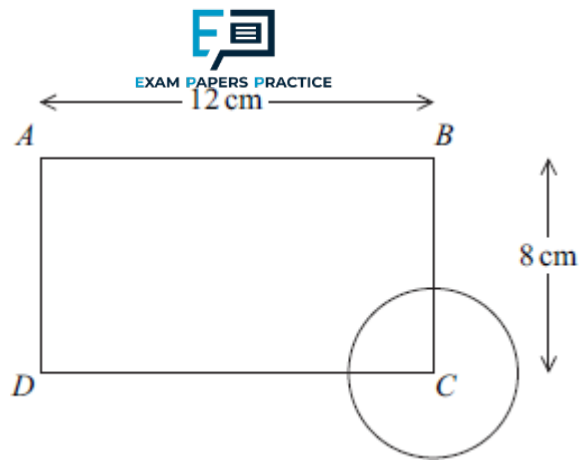
(4)

(Total 8 marks)

Q3.

A uniform rectangular lamina $ABCD$, of mass 1.6 kg , has side AB of length 12 cm and side BC of length 8 cm .

To create a logo, a uniform circular lamina, of mass 0.4 kg , is attached. The centre of the circular lamina is at the point C , as shown in the diagram.

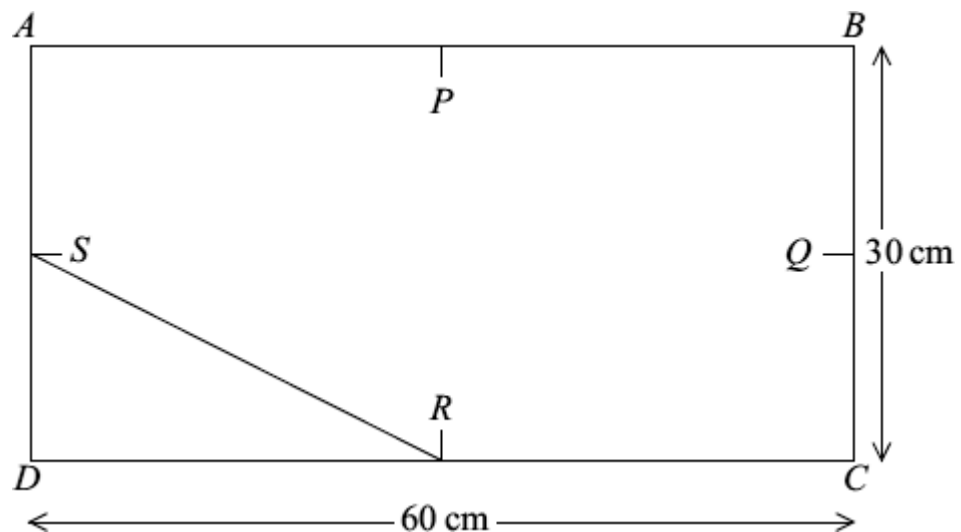


- (a) Find the distance of the centre of mass of the logo:
- (i) from the line AB ; (3)
- (ii) from the line AD . (3)
- (b) The logo is suspended in equilibrium, with AB horizontal, by two vertical strings. One string is attached at the point A and the other string is attached at the point B . Find the tension in each of the two strings. (5)
- (Total 11 marks)

Q4.

A uniform rectangular lamina $ABCD$ has a mass of 5 kg. The side AB has length 60 cm and the side BC has length 30 cm. The points P , Q , R and S are the mid-points of the sides, as shown in the diagram below.

A uniform triangular lamina SRD , of mass 4 kg, is fixed to the rectangular lamina to form a shop sign. The centre of mass of the triangular lamina SRD is 10 cm from the side AD and 5 cm from the side DC .



- (a) Find the distance of the centre of mass of the shop sign from AD .

(3)

- (b) Find the distance of the centre of mass of the shop sign from AB .

(3)

- (c) The shop sign is freely suspended from P .

Find the angle between AB and the horizontal when the shop sign is in equilibrium.

(4)

- (d) To ensure that the side AB is horizontal when the shop sign is freely suspended from point P , a particle of mass m kg is attached to the shop sign at point B .

Calculate m .

(3)

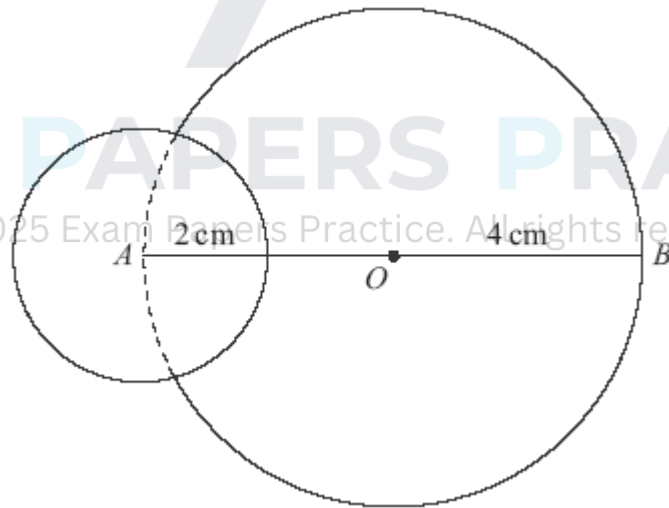
- (e) Explain how you have used the fact that the rectangular lamina $ABCD$ is uniform in your solution to this question.

(1)

(Total 14 marks)

Q5.

A uniform circular lamina, of radius 4 cm and mass 0.4 kg, has a centre O , and AB is a diameter. To create a medal, a smaller uniform circular lamina, of radius 2 cm and mass 0.1 kg, is attached so that the centre of the smaller lamina is at the point A , as shown in the diagram.



- (a) Explain why the centre of mass of the medal is on the line AB .

(1)

- (b) Find the distance of the centre of mass of the medal from the point B .

(3)

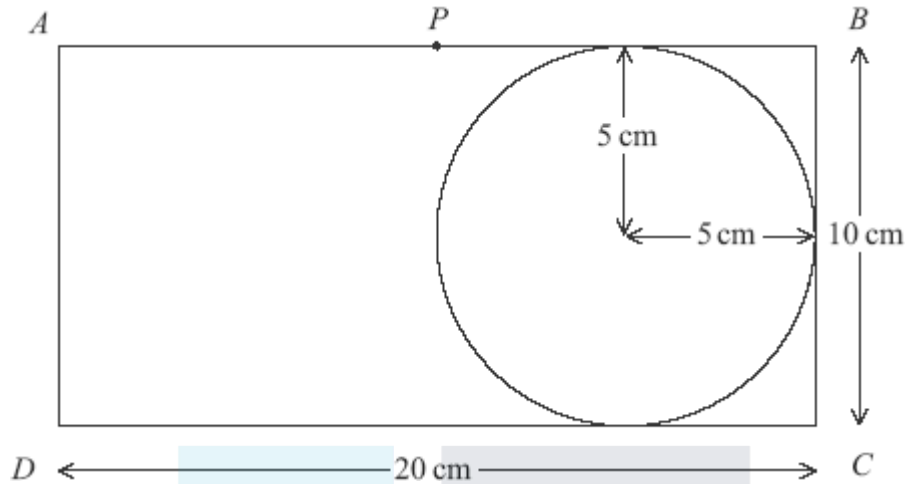
(Total 4 marks)

Q6.

A uniform rectangular lamina $ABCD$ has a mass of 8 kg. The side AB has length 20 cm,

the side BC has length 10 cm, and P is the mid-point of AB .

A uniform circular lamina, of mass 2 kg and radius 5 cm, is fixed to the rectangular lamina to form a sign. The centre of the circular lamina is 5 cm from each of AB and BC , as shown in the diagram.



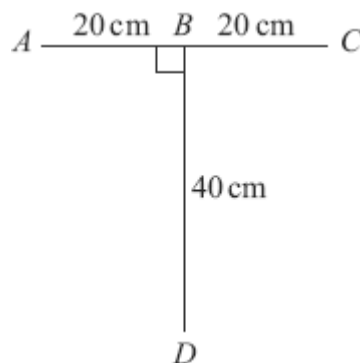
- (a) Find the distance of the centre of mass of the sign from AD . (3)
- (b) Write down the distance of the centre of mass of the sign from AB . (1)
- (c) The sign is freely suspended from P .
Find the angle between AD and the vertical when the sign is in equilibrium. (4)
- (d) Explain how you have used the fact that each lamina is uniform in your solution to this question. (1)

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(Total 9 marks)

Q7.

Two identical uniform rods, AC and BD , are rigidly joined together to form a letter T, as shown in the diagram. The two rods are perpendicular.



- (a) Explain why the centre of mass of the letter T lies on BD .

(1)

- (b) Find the distance of the centre of mass of the letter T from AC .

(3)

- (c) The letter T is freely suspended from A .

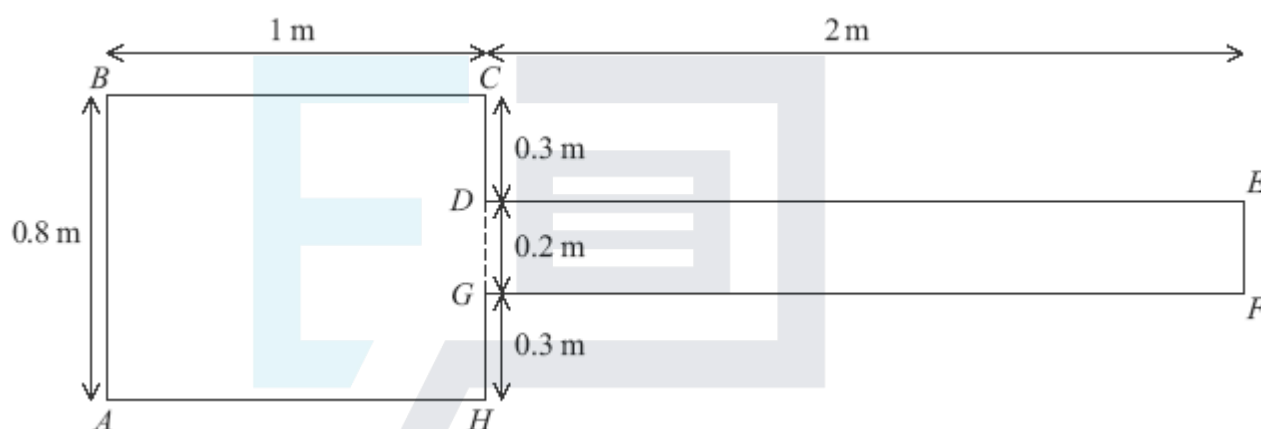
Find, to the nearest degree, the angle between AC and the horizontal when the letter T hangs in equilibrium.

(3)

(Total 7 marks)

Q8.

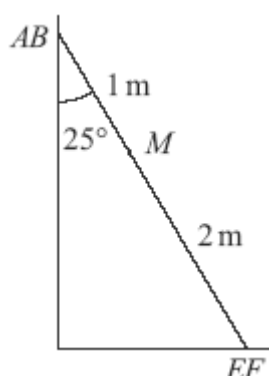
A sign advertising a gym consists of two rectangles $ABCH$ and $DEFG$ fixed rigidly together. The sign can be modelled as a uniform lamina, as shown in the diagram.



- (a) The centre of mass of the sign is at the point M . Show that M lies on the line CH .

(4)

- (b) The sign is placed with its side EF on rough horizontal ground and its side AB against a smooth vertical wall. The sign rests in equilibrium at an angle of 25° with the **vertical**, as shown in the diagram.



The weight of the sign is 90 newtons.

- (i) By taking moments, show that the normal reaction force between the sign and the wall is $60 \tan 25^\circ$ newtons.

(3)

- (ii) The coefficient of friction between the sign and the ground is μ .

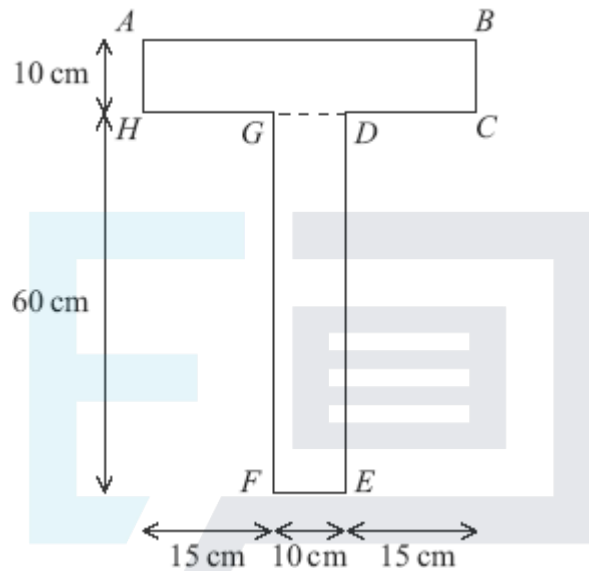
Show that $\mu \geq \frac{2}{3} \tan 25^\circ$.

(4)

(Total 11 marks)

Q9.

A uniform T-shaped lamina is formed by rigidly joining two rectangles $ABCH$ and $DEFG$, as shown in the diagram.



- (a) Show that the centre of mass of the lamina is 26 cm from the edge AB .

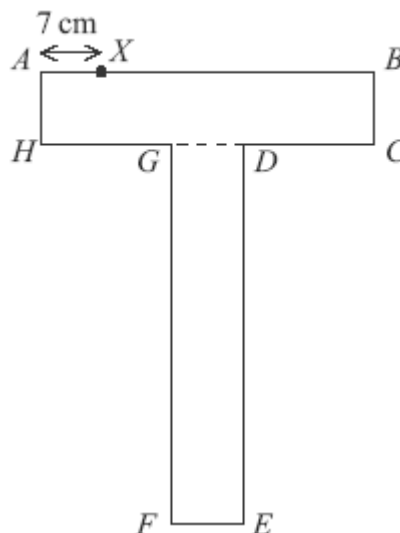
(4)

- (b) Explain why the centre of mass of the lamina is 5 cm from the edge GF .

(1)

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- (c) The point X is on the edge AB and is 7 cm from A , as shown in the diagram below.



The lamina is freely suspended from X and hangs in equilibrium.

Find the angle between the edge AB and the vertical, giving your answer to the nearest degree.

(4)

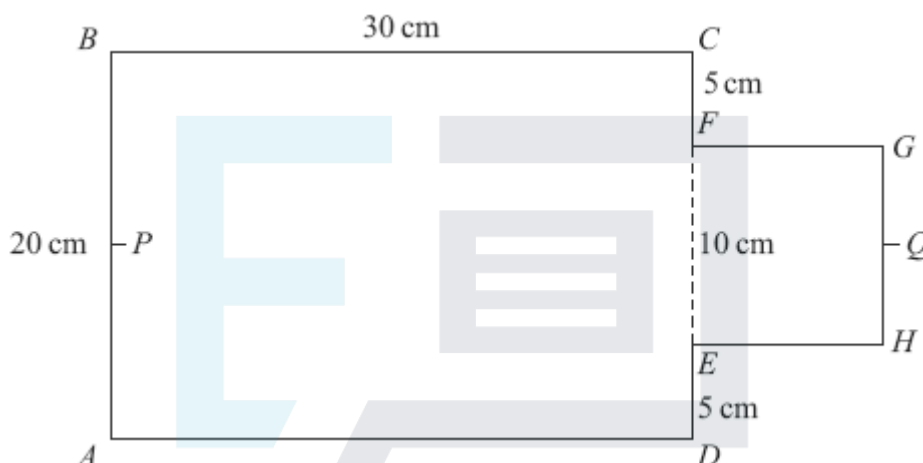
(Total 9 marks)

Q10.

A uniform lamina is in the shape of a rectangle $ABCD$ and a square $EFGH$, as shown in the diagram.

The length AB is 20 cm, the length BC is 30 cm, the length DE is 5 cm and the length EF is 10 cm.

The point P is the midpoint of AB and the point Q is the midpoint of HG .



- (a) Explain why the centre of mass of the lamina lies on PQ .

(1)

- (b) Find the distance of the centre of mass of the lamina from AB .

(4)

- (c) The lamina is freely suspended from A .

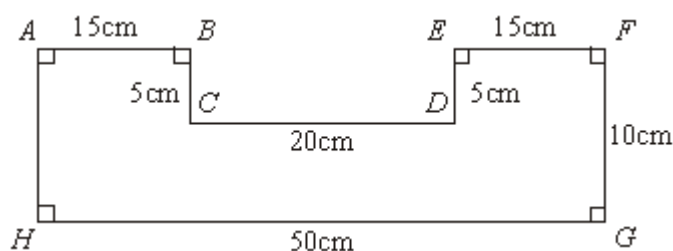
Find, to the nearest degree, the angle between AD and the vertical when the lamina is in equilibrium.

(4)

(Total 9 marks)

Q11.

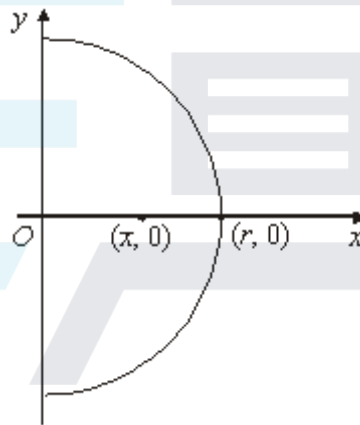
The diagram shows a uniform lamina $ABCDEFGH$.



- (a) Explain why the centre of mass is 25 cm from AH . (1)
- (b) Show that the centre of mass is 4.375 cm from HG . (4)
- (c) The lamina is freely suspended from A . Find the angle between AB and the vertical when the lamina is in equilibrium. (4)
- (d) Explain, briefly, how you have used the fact that the lamina is uniform. (1)
- (Total 10 marks)

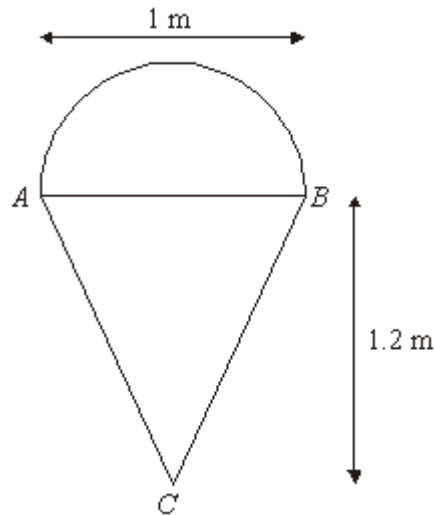
Q12.

- (a) A uniform semicircular lamina of radius r has its centre at the origin and its axis of symmetry along Ox . The position of its centre of mass has coordinates $(\bar{x}, 0)$.



- (i) Show that $\frac{1}{2} \pi r^2 \bar{x} = \int_0^r 2x \sqrt{r^2 - x^2} dx$. (4)
- (ii) Hence prove that $\bar{x} = \frac{4r}{3\pi}$. (3)

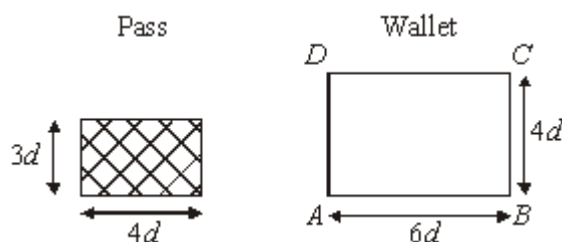
- (b) The diagram below shows a uniform lamina, used as an advertising feature in a local supermarket to promote ice cream. It consists of a semicircle, of diameter $AB = 1$ m, and an isosceles triangle ABC , where C is at a distance 1.2 m from AB .



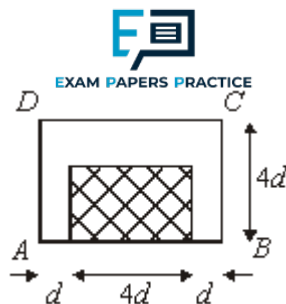
- (i) State the distance of the centre of mass of the **triangle** from C . (1)
- (ii) Show that the distance of the centre of mass of the **semicircle** from C is approximately 1.41 m . (1)
- (iii) Find the distance of the centre of mass of the complete lamina from C . (4)
- (c) The lamina is freely suspended from A . Find the angle that AB makes with the vertical through A , giving your answer to the nearest degree. (3)
- (Total 16 marks)

Q13.

A security pass, of length $4d$ and width $3d$, consists of a rectangular piece of plastic. The pass can be carried in a transparent plastic wallet, $ABCD$, where $AB = 6d$ and $BC = 4d$. The pass and the wallet, which may be modelled as uniform laminas, are shown in the diagrams.



The pass is placed inside the wallet with its longer edge along AB and its centre equidistant from BC and AD , as shown in the diagram below.



The mass of the pass is $4M$ and the mass of the wallet is M .

- (a) Show that the distance of the centre of mass of the pass and wallet above AB is $1.6d$.

(4)

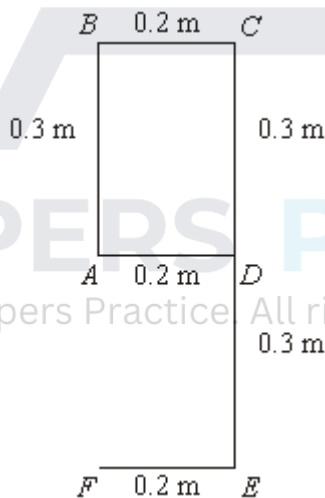
- (b) The wallet, with the pass still inside and in the position shown in the diagram, is suspended freely from the point C . Find the angle, in degrees, that the edge CD makes with the vertical.

(4)

(Total 8 marks)

Q14.

As part of an advertising display, a thin uniform rod of length 1.5 m is bent to form a figure . The diagram shows the figure in which $ABCD$ is a rectangle and angle $ADE = \text{angle } DEF = 90^\circ$:



- (a) The mass of BC is $2M$.

Express the total mass of the figure as a multiple of M .

(1)

- (b) (i) Show that the centre of mass of the figure is at a distance of 0.27 m from BC .

(4)

- (ii) Find the distance of the centre of mass of the figure from CE .

(3)

- (c) When freely suspended from the point C , the side CE of the figure makes an angle

θ with the vertical.

Determine the value of θ , giving your answer in degrees to one decimal place.

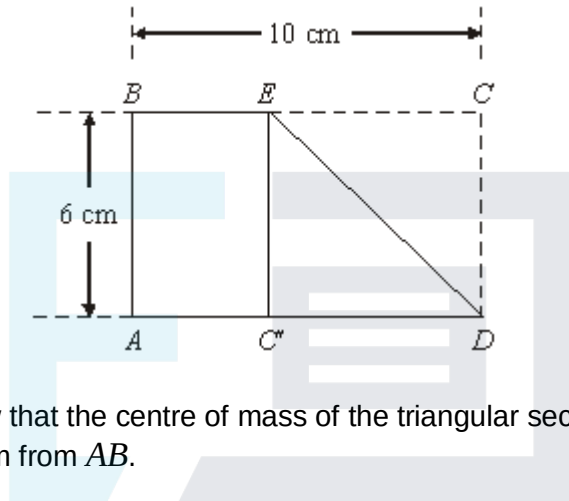
(3)

(Total 11 marks)

Q15.

A uniform rectangular lamina $ABCD$ has sides $AB = 6$ cm and $BC = 10$ cm.

The section ECD is folded along the line DE so that DC lies along DA and the point C is at C' , as shown in the diagram.



- (a) (i) Show that the centre of mass of the triangular section $EC'D$ of the new body is 6 cm from AB . (2)
- (ii) Hence show that the centre of mass of the new body is 4.4 cm from AB . (2)
- (iii) Find the distance of the centre of mass of the new body from AD . (2)

- (b) The body is now freely suspended from B and is then rotated in the vertical plane of the body to the position where AD is horizontal, as in the diagram above. It is held in equilibrium in this position by a couple of magnitude K .

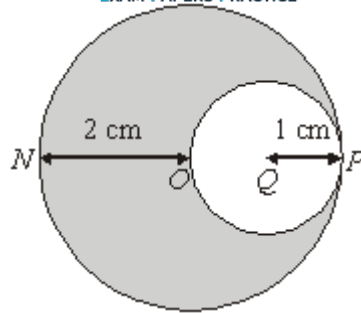
Given that the weight of the body is 2 N, find the value of K .

(2)

(Total 8 marks)

Q16.

The diagram shows an earring which can be modelled as part of a uniform circular lamina. It is made by removing a small circle from a larger circle, as shown in the diagram.



The larger circle has centre O and diameter NP . The radius of this circle is 2 cm. The smaller circle has centre Q and diameter OP . The radius of this circle is 1 cm.

Determine the distance of the centre of mass of the earring from N .

(Total 5 marks)

Q17.



A garden spade can be modelled by a uniform rod, AB , together with a uniform rectangular plate $CDEF$. The plate is rigidly attached to the rod at B , which is the midpoint of CF , as shown in the diagram.

The rod is of length 0.8 m and mass 0.25 kg. The plate is such that $CD = FE = 0.3$ m, $CF = DE = 0.2$ m and its mass is 1 kg.

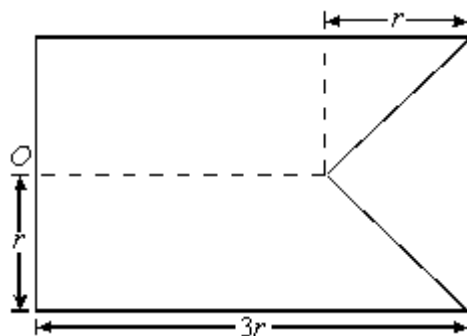
Find the distance of the centre of mass of the spade from A .

(Total 4 marks)

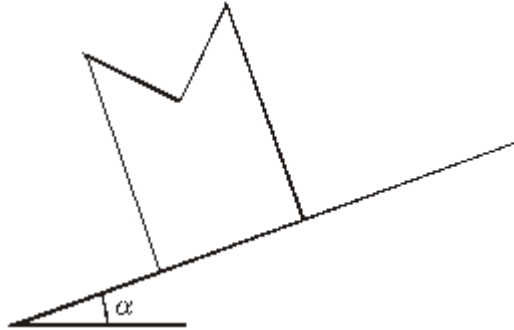
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Q18.

A solid, uniform cylinder, of radius r and length $3r$, has a cone of radius r and height r drilled out of one end. The diagram shows a cross section along the axis of symmetry of the body. Point O is the centre of the plane circular surface of the body.



- (a) Show that the centre of mass of the body is $\frac{43r}{32}$ from point O . (6)
- (b) The body is placed with its plane surface on an adjustable rough slope as shown. The angle α which the plane makes with the horizontal is gradually increased from zero.

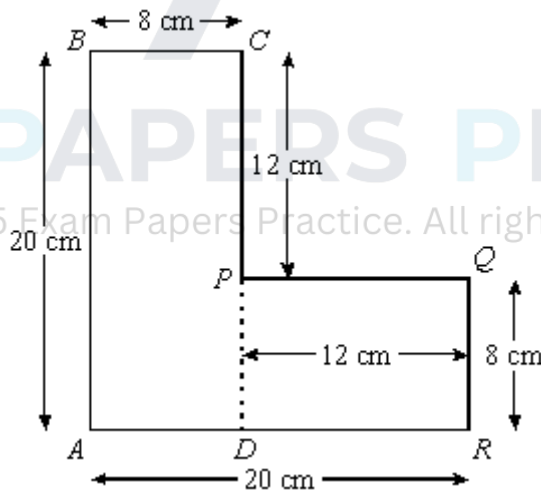


Assuming that the body does **not** slide, find, to the nearest degree, the value of angle α when the body just begins to topple.

(3)
(Total 9 marks)

Q19.

The diagram shows a uniform lamina, which consists of two rectangles $ABCD$ and $DPQR$.



The dimensions are such that:

$$\begin{aligned}
 DR &= PQ = CP = 12 \text{ cm}; \\
 BC &= QR = 8 \text{ cm}; \\
 AB &= AR = 20 \text{ cm}.
 \end{aligned}$$

- (a) Explain why the centre of mass of the lamina must lie on the line AP . (1)
- (b) Find the distance of the centre of mass of the lamina from AB .

(4)

- (c) The lamina is freely suspended from B .

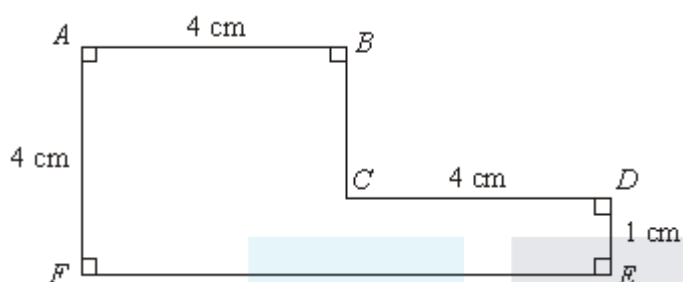
Find, to the nearest degree, the angle that AB makes with the vertical through B .

(4)

(Total 9 marks)

Q20.

The diagram shows a uniform lamina.



- (a) Show that the centre of mass of the lamina is 2.8 cm from AF .

(4)

- (b) Find the distance of the centre of mass from AB .

(3)

- (c) The lamina is suspended in equilibrium from the corner A . Find the angle between AB and the vertical.

(3)

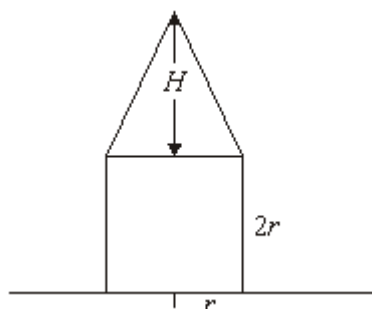
(Total 10 marks)

Q21.

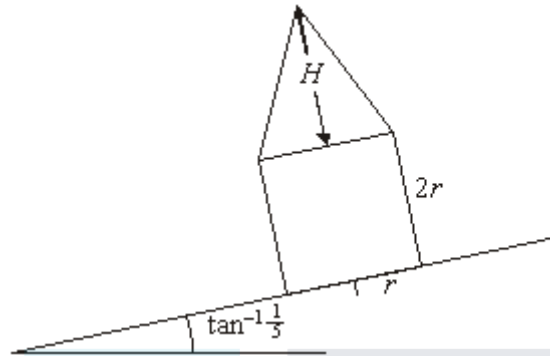
The diagram shows a cross-section of a child's toy, which consists of a uniform solid cylinder and a uniform solid right circular cone. One plane face of the cylinder is fixed to the plane face of the cone.

The cylinder has density ρ , radius r and height $2r$.

The cone has density 3ρ , height H and radius r .



- (a) Show that the centre of mass of the toy is $\frac{8r^2 + 8rH + H^2}{4(2r + H)}$ from its base. (6)
- (b) The toy is now placed on a rough plane inclined at an angle of $\tan^{-1} \frac{1}{5}$ to the horizontal.

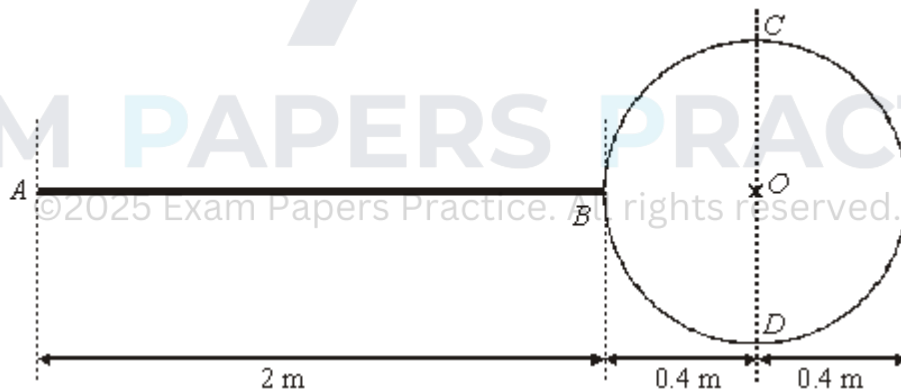


Show that the maximum value of H if the toy is not to topple is $(6 + 2\sqrt{17})r$.

(5)
(Total 11 marks)

Q22.

A uniform circular disc is attached to one end of a uniform pole to form the body shown in the diagram below.



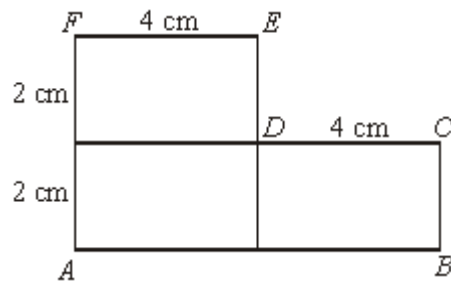
The pole AB has length 2 metres and mass 6 kg.
 The disc, centre O , has radius 0.4 metres and mass 2 kg.
 The points A , B and O lie on a straight line.

- (a) Show that the centre of mass of the body is 1.35 metres from A . (3)
- (b) The diameter, CD , of the disc is perpendicular to OA . The body is suspended from the point C . Find the angle between the pole and the vertical when the body hangs at rest in equilibrium. (4)

(Total 7 marks)

Q23.

A uniform lamina is formed by joining together three identical rectangles, each with dimensions 2 cm by 4 cm. The lamina is shown in the diagram below.



- (a) Show that the distance of the centre of mass of the lamina from AF is $\frac{10}{3}$ cm. (4)
- (b) Find the distance of the centre of mass of the lamina from AB . (3)
- (c) The lamina is freely suspended in equilibrium from the corner A . Find the angle between AB and the vertical. (4)

(Total 11 marks)